W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Pentester Name (Cybersecurity Professional)	**Md. Shihab Shahriar Rashu**
Program / Batch	B084-Networkwalks
Date	12 November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (networkwalks.com — secured written permission) 2. Local Area Network (10.0.0.0/24 Subnet)
Permission Secured from Client?	Yes (Authorized Lab & Controlled Environment)
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3–5: In Progress

1. LIABILITY DISCLAIMER

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2. INTRODUCTION

This formal penetration testing report covers footprinting the `networkwalks.com` domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they demonstrate how an attacker moves systematically from gathering public information to mapping live hosts on a network. It constitutes the Week 2 deliverable of my ongoing cybersecurity internship program at Networkwalks.

All commands were executed in Kali Linux (footprinting) and on a workstation with Zenmap installed (scanning). Every step below includes the exact command used, the result observed, screenshot evidence, and a concise technical note on why the finding matters from a security assessment perspective.

3. TOOLS USED

Tool	Operational Purpose
Kali Linux	Primary penetration testing operating system used for reconnaissance activities.
WHOIS	Find domain registration details (owner, dates, name servers, registrar information).
whatweb	Fingerprint web technologies (server, CMS, plugins, versions, IP address).
nslookup	Resolve the domain name to its IP address using DNS queries.
curl -I	Inspect and read the HTTP/HTTPS response headers of the target website.
wafw00f	Detect whether a Web Application Firewall (WAF) protects the site.
dnsrecon	Enumerate comprehensive DNS records (SOA, NS, MX, SPF, TXT, SRV, BIND version).
Zenmap (Nmap GUI)	Scan the local subnet to discover live hosts, IP addresses, latency, and MAC addresses.
Linux Terminal / CLI	Execution environment for command-line information gathering utilities.

4. ACTIVITIES PERFORMED

4.1 Footprinting & Reconnaissance

I performed active and passive reconnaissance against the `networkwalks.com` domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon. Each tool was systematically utilized to gather targeted technical information regarding domain infrastructure, web server technologies, defensive controls, and DNS service profiles.

- **WHOIS:** Obtained publicly available domain registration data. The query identified the registrar (GoDaddy.com, LLC with IANA ID 146), registration timeline (created 2019-11-06, expiry 2027-11-06), privacy guard protection (Domains By Proxy, LLC), and primary authoritative name servers (`NS6135.HOSTGATOR.COM` and `NS6136.HOSTGATOR.COM`).
- **WhatWeb:** Fingerprinted technologies operating on the web application. Discovered Apache web server, Bootstrap 7.1.4, jQuery 3.7.1, WordPress 7.1.4 CMS, and an active plugin (WordPress Download Manager 3.3.76), operating on IP `192.232.216.135`.
- **Nslookup:** Resolved the domain name against Google Public DNS (`8.8.8.8#53`). The non-authoritative response mapped the domain directly to host IP address `192.232.216.135`.
- **Curl (`curl -I`):** Inspected HTTP response headers against `https://networkwalks.com`. The returned HTTP/2 200 response exposed Apache server software, caching headers (`x-nginx-cache: WordPress`, `x-endurance-cache-level: 0`), session cookies (`__wpdm_client`), and exposed the WordPress REST API endpoint (`/wp-json/`).
- **Wafw00f:** Analyzed perimeter defense and confirmed that the target web application is actively protected behind a **ModSecurity (SpiderLabs) Web Application Firewall**.
- **DNSRecon:** Enumerated complete DNS infrastructure: SOA at `ns6135.hostgator.com` (50.87.144.87), authoritative Name Servers (NS6135 and NS6136), Mail Exchanger (`mail.networkwalks.com` at 192.232.216.135), host address (A record 192.232.216.135), BIND software version disclosure (`"9.16.23-RH"`), SPF/TXT verification records, and eight autodiscover SRV records routing to `cpanelemaildiscovery.cpanel.net`.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery and host enumeration on my local network. The objective was to discover active hosts across the local subnet, record their responsive IP and MAC addresses, measure latency, and map the physical and logical topology.

I executed a Ping Scan profile (`nmap -sn 10.0.0.0/24`) across the `10.0.0.0/24` subnet. The scan successfully probed 256 IP addresses in 3.34 seconds and identified three live hosts:

- **10.0.0.1:** Host is up (0.0094s latency) | MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
- **10.0.0.2:** Host is up (Local scanning interface / gateway)
- **10.0.0.3:** Host is up (0.0049s latency) | MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)

Following host discovery, I accessed the Topology section in Zenmap, toggled the graphic visualizer, and generated a ring topology showing host positions relative to localhost, exporting the graphical visual evidence.

5. RISK ANALYSIS / IMPACT

Based on the data collected during the footprinting and network scanning phases, I evaluated the potential security implications. The table below outlines the identified findings, evidence, attacker perspective, and associated risk levels.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology & plugin information exposed	WhatWeb revealed WordPress 7.1.4, WP Download Manager 3.3.76, jQuery 3.7.1	Attackers may use exposed technology/version information to research targeted CVEs and public exploit code.	• Medium
2	Server IP address identifiable	Nslookup & DNSRecon resolved domain and MX record to 192.232.216.135	Provides direct network host location, allowing potential bypass of DNS-level proxy protections.	• Low
3	HTTP technical headers & REST API exposed	Curl returned Apache headers, cache info, and exposed /wp-json/ link endpoint	Assists attackers in automated fingerprinting and aggressive API user enumeration.	• Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs) WAF in 2 requests	Reveals defensive architecture, allowing adversaries to craft specialized WAF-evasion payloads.	• Low
5	DNS infrastructure & BIND version exposed	DNSRecon revealed BIND '9.16.23-RH', mail records, and SRV entries	Software disclosures allow targeting version-specific daemon vulnerabilities; mail records aid spear-phishing.	• Medium
6	Multiple live hosts visible on local network	Zenmap ping scan identified 3 live hosts (10.0.0.1, 10.0.0.2, 10.0.0.3) with QEMU MAC addresses	Unknown, unsegmented, or unauthorized virtual appliances may potentially be present on the internal network.	• Medium

Risk level key: ● Critical ● Medium ● Low
The risks detailed above represent architectural observations and reconnaissance exposures gathered during non-intrusive footprinting and network discovery phases. No vulnerability exploitation or payload delivery was performed. While technology disclosure lowers the reconnaissance barrier for adversaries, it does not imply compromise without active exploitable vulnerabilities.

6. RECOMMENDATIONS

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information:** Organizations should regularly review what information about their web technologies, CMS, and plugins is publicly visible. Suppress version strings in CMS generator tags and script sources.
2. **Keep software updated:** CMS platforms, plugins, and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers:** HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed. Obscure detailed Apache banners using `ServerTokens Prod` and remove unnecessary caching headers.
4. **Review DNS records regularly:** DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Obscure DNS server version:** Suppress the authoritative BIND version string disclosure (`9.16.23-RH`) by defining `version none;` or custom responses in the nameserver configuration.
6. **Properly configure and monitor the WAF:** Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks, and ensure OWASP Core Rule Sets (CRS) are kept up to date.
7. **Perform regular internal network discovery:** Organizations should periodically scan their own networks to identify active devices and eliminate rogue virtual systems.
8. **Investigate unknown devices:** Any unexpected device discovered during network scanning should be investigated and verified.
9. **Maintain network documentation:** Network topology and device information should be documented and updated regularly.
10. **Perform security testing with authorization:** Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. CONCLUSION

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering Phase 1 (Footprinting & Reconnaissance) and Phase 2 (Scanning & Network Discovery).

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk. Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. EVIDENCES COLLECTED

Below is the visual evidence collected from the terminal sessions and GUI scanning interfaces during the execution of W2-PM1 and W2-PM5.

Figure 1: WHOIS Domain Information & Nameserver Output **Tool: whois**



Figure 2: Web Technology & CMS Fingerprinting **Tool: whatweb**

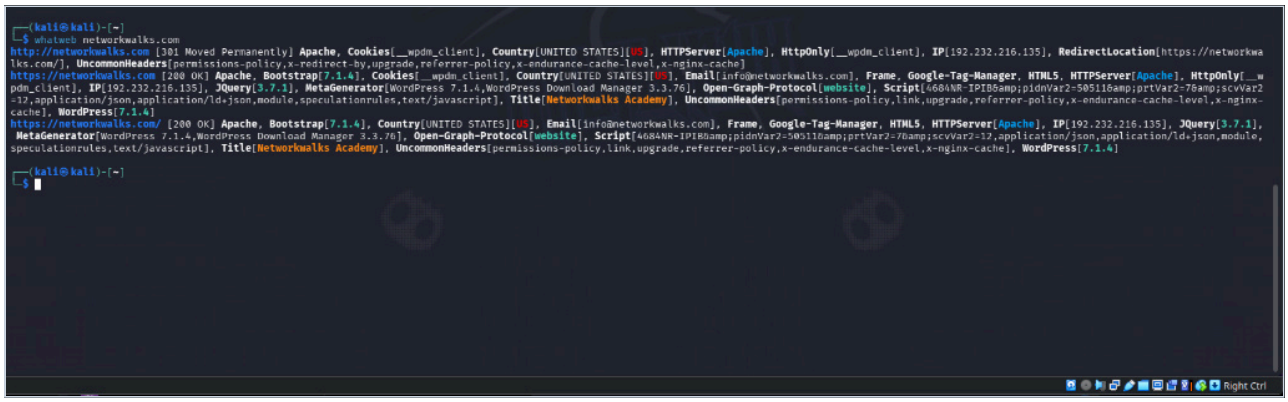


Figure 3: DNS A-Record Resolution to Host IP **Tool: nslookup**

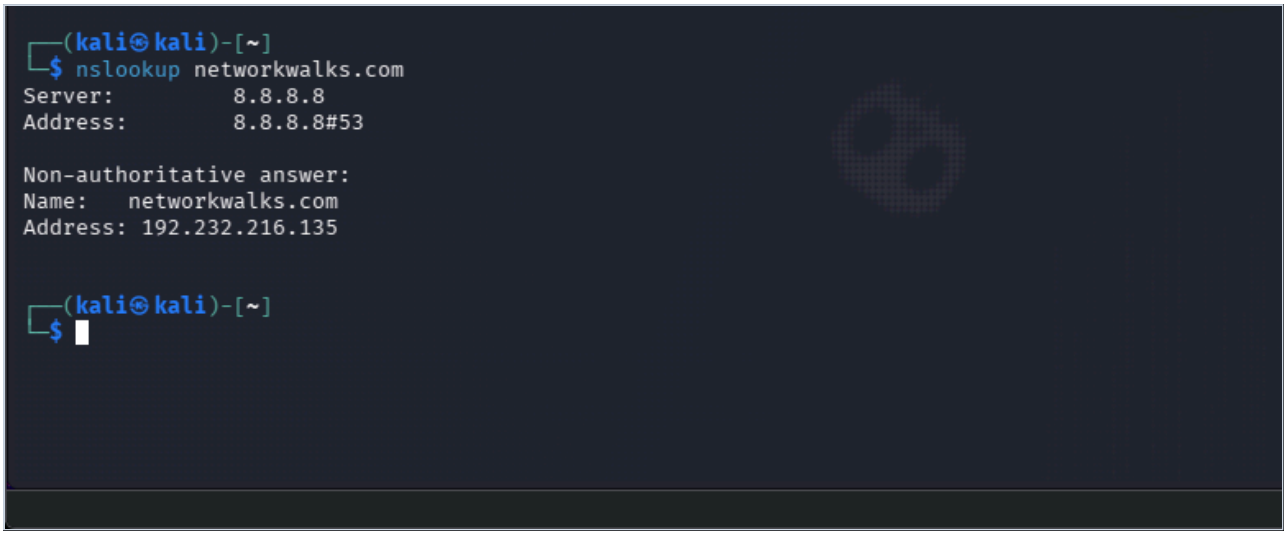


Figure 4: HTTP Response Headers & API Endpoint Inspection **Tool: curl -I**



Figure 5: Web Application Firewall Detection (ModSecurity) **Tool: wafw00f**

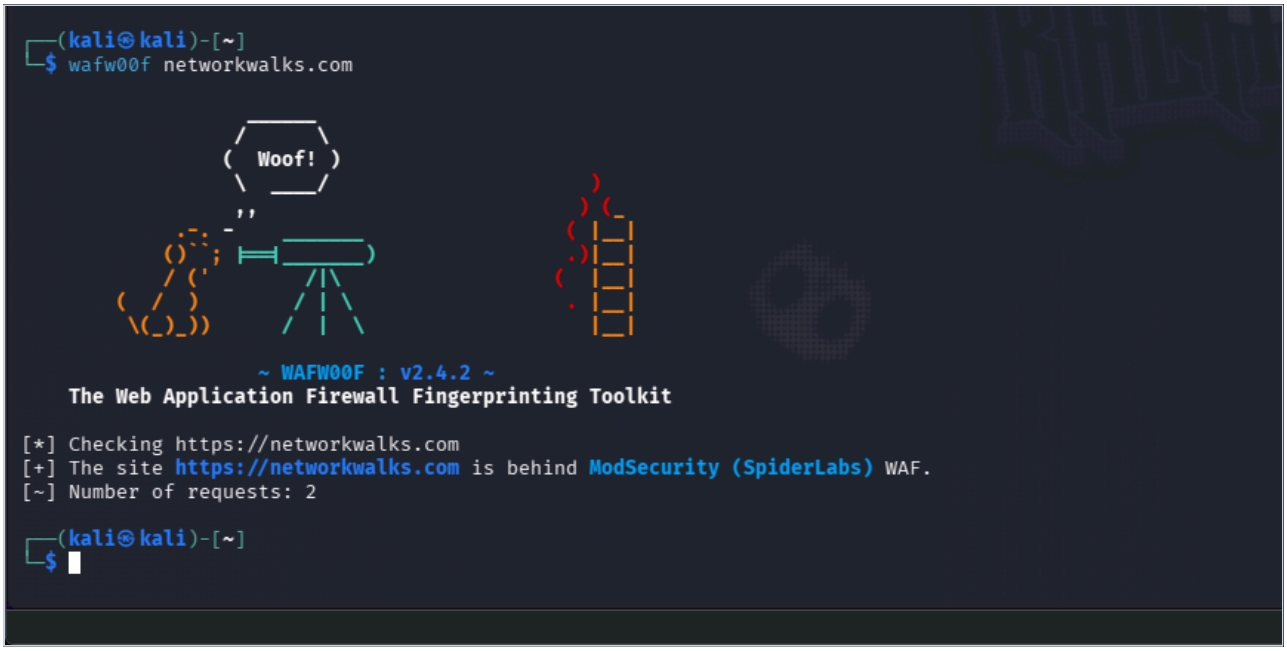


Figure 6: Complete DNS & SRV Record Enumeration **Tool: dnsrecon**

```
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-11T13:34:28.965769-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-11T13:34:28.966139-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-11T13:34:29.623629-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-11T13:34:30.012627-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-11T13:34:30.733082-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-11T13:34:31.535321-0500 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-11T13:34:31.535578-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-11T13:34:32.129955-0500 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-11T13:34:33.053369-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-11T13:34:33.556497-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-11T13:34:34.652277-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-11T13:34:34.652518-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-11T13:34:35.459296-0500 INFO Enumerating SRV Records
2026-11-11T13:34:38.983813-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-11T13:34:38.984099-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-11T13:34:38.984223-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-11T13:34:38.984415-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-11T13:34:38.984534-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-11T13:34:38.984612-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-11T13:34:38.984678-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-11T13:34:38.984755-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-11T13:34:38.985586-0500 INFO 8 Records Found
2026-11-11T13:34:38.985831-0500 INFO Completed enumeration for domain: networkwalks.com

┌──(kali㉿kali)-[~]
└─$
```

Figure 7: Zenmap Local Subnet Ping Scan Output (10.0.0.0/24) **Tool: Zenmap / Nmap**

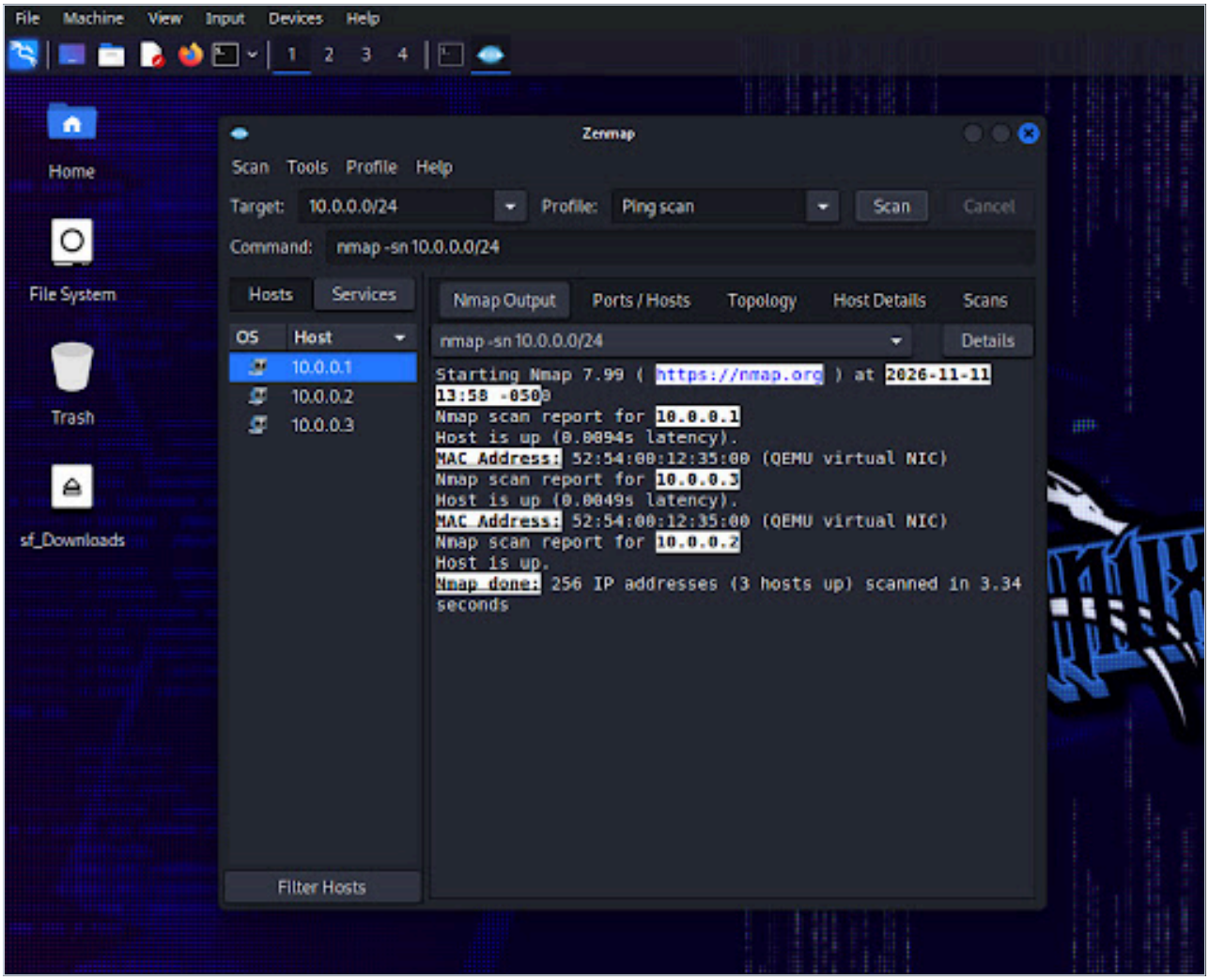
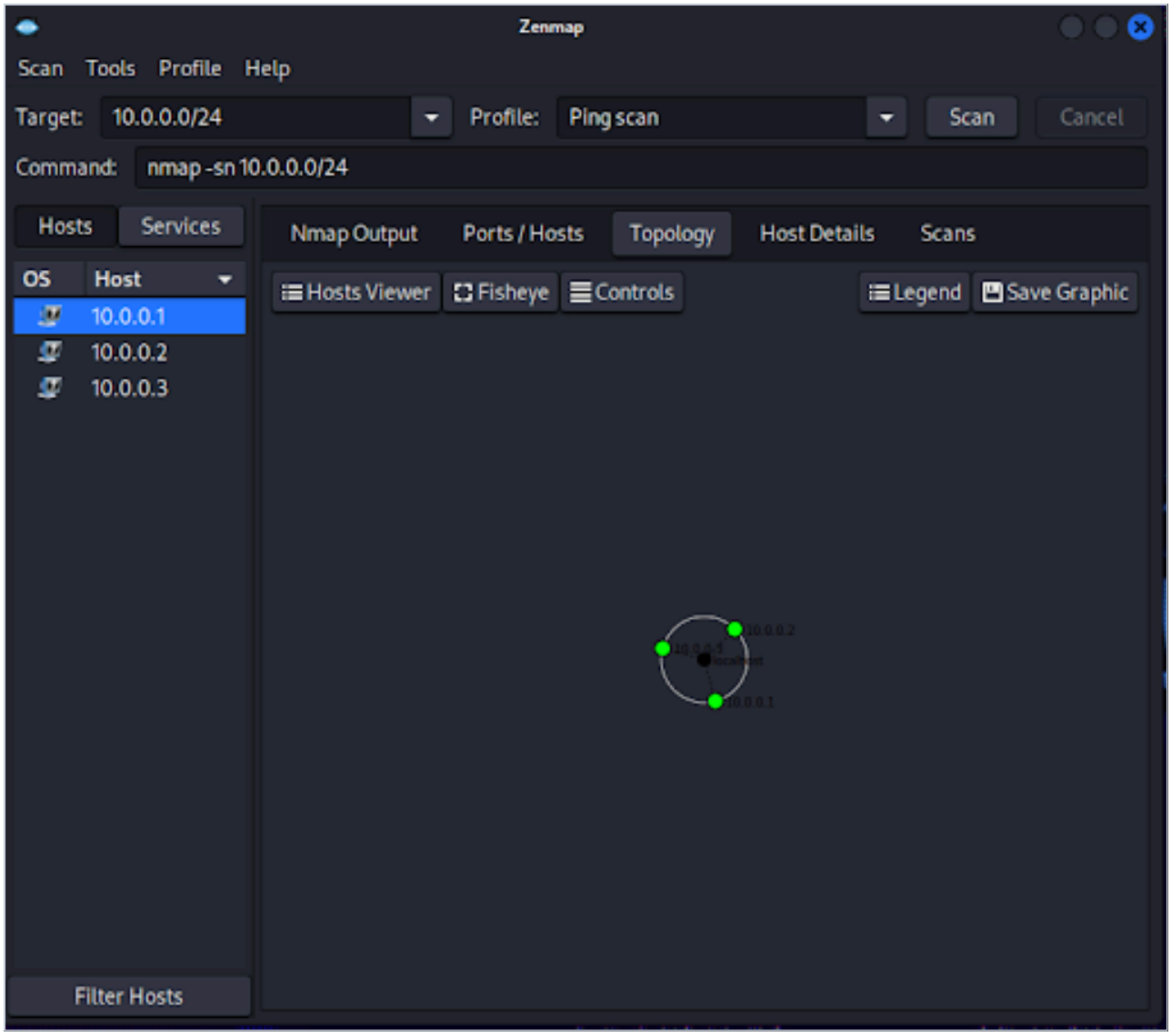


Figure 8: Zenmap Network Topology Visualization Graphic **Tool: Zenmap Topology Viewer**



- End of Technical Deliverables -

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Project Deliverables:
Program: Cybersecurity Program at Networkwalks (Week 02)
Modules: W2-PM1 & W2-PM5 | Date: 12 November 2026